Energy & Buildings 170 (2018) 107–114

Contents lists available at ScienceDirect

Energy & Buildings

journal homepage: www.elsevier.com/locate/enbuild

Determinants of high electricity use and high energy consumption for space and water heating in European social housing: Socio-demographic and building characteristics

Stavroula Karatasou[a,*], Marina Laskari[a], Mat Santamouris[b]

[a] Group Building of Environmental Physics, University of Athens, Building 5, Physics University Campus, Athens 15784, Greece
[b] High Performance Architecture, School of Built Environment, University of New South Wales, Sydney, NSW, Australia

ARTICLE INFO

Article history:
Received 29 December 2017
Revised 20 March 2018
Accepted 12 April 2018
Available online 18 April 2018

ABSTRACT

This paper examines the extent to which different household socio-demographic and building characteristics contribute to high energy use in Social Housing. Using data from the sample created in the framework of the EU funded ICE-WISH project (social housing pilots in ten European countries, of around 25 dwellings each), an odds ratio analysis was performed to examine the effect of socio-demographic and building's structural determinants on high residential energy consumption. The analysis is performed separately for electricity use for lighting and appliances and for energy consumption for space and water heating. Results indicate that high electricity consumption in social housing is related to both socio-demographic characteristics of the occupants and characteristics of the buildings in which they live (OR = 5.6 for households with three occupants compared to single households and OR = 3 for households living in houses greater than 80 m^2), whereas high energy consumption for space and water heating is only related to building characteristics (OR = 1.15 for households with three occupants compared to single households and OR = 4.77 for households living in houses greater than 80 m^2). The outcomes of this paper call for future energy policy to intelligently incorporate traditional and behavior-based energy efficiency interventions for reducing residential energy consumption and CO2 emissions.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

Households consume a great amount of energy and thus contribute considerably to greenhouse gas emissions, through the everyday energy-related behaviors and the use of energy-intensive products and services by occupants [1]. In the EU, the residential building sector is responsible for about 26.8% of total energy consumption [2], while in the US the residential sector consumes 22% of the total energy [3].

In Europe, during the years 2005–2009, energy efficiency in the household sector increased at an annual average rate of 1.4% per year, driven by the implementation of EPBD and the diffusion of more efficient buildings, space heating technologies and electrical appliances. Over the same period, electricity consumption grew faster, mainly due to the increased number of owned electrical appliances and household size [4].

Nevertheless, variation in household energy consumption remains large, even for buildings with no physical differences in built structures. In order to be able to harness the great potential for energy conservation and emissions reduction in residential buildings it is important to gain a sound understanding of the determinants that drive high total energy consumption first [5,6]. The research to date indicates that residential energy consumption is determined by many factors, including climate, building characteristics, demographics and socio-economics, socio-psychological and behavior aspects. However, because of lack of easily accessible and high resolution consumption data, determinants of energy use and energy-related behaviors have not been extensively investigated [6,7]. Most of the existing studies focus on electricity consumption, [8 and references therein], and analyze constrainedly a partial set of residential electricity consumption determinants [9,10]. Moreover, limited studies examine energy use for space and water heating (SWH) and the effect of thermal characteristics of the building stock from a statistical perspective [5,11].

This study aims to gain a greater insight into the drivers of high energy consumption in social housing in Europe, by examining the relative effect of building and socio-demographic characteristics. Our analysis uses measured utility data for electricity consumption and consumption for space and water heating, as well as questionnaire survey data for building and socio-demographic

* Corresponding author.
E-mail address: skarat@phys.uoa.gr (S. Karatasou).

https://doi.org/10.1016/j.enbuild.2018.04.019
0378-7788/© 2018 Elsevier B.V. All rights reserved.

characteristics. The paper is structured as follows: Section 2 provides a literature review while Sections 3 and 4 present the analysis methods and data used in this study and the results of the statistical analysis, respectively. Section 5 discusses the results of the study and Section 6 provides the conclusions.

2. State of the art

Previous literature has demonstrated, through numerous empirically based studies, that there are several factors affecting residential energy consumption. Most of the studies focus on electricity consumption [8]. In a similar way, few articles are available discussing factors affecting the energy consumption for space and water heating from a statistical point of view [5,11].

2.1. Impact of building characteristics

In relation to the building characteristics that influence the energy consumption, the most commonly used explanatory variables in previous studies include: the building type and size, the year of construction and the construction materials.

Residential energy consumption varies across the different housing types [12,13]. Detached single-family units consume more energy for heating and cooling than households in multifamily housing units [14]. Apartments use less energy than detached or semidetached housing structures, as they usually consist of smaller spaces and their envelope is less exposed to external conditions [15].

The size of residential buildings is positively correlated with the energy consumption, and in general larger residential buildings, as measured by number of rooms, number of floors or floor area, consume more energy than smaller residential buildings [14–21]. Energy intensity (kWh/m^2) is used as an indicator for analyzing electricity consumption [11,22–27]. However, the correlation coefficient of energy use per unit area with respect to floor area is negative [28], suggesting that dwellings become more efficient as they become larger.

Regarding the year of construction, findings vary across studies. In general, older buildings use more energy than newer buildings [18,20,28–29]. Similar results were also found by Guerra Santin et al. for the Dutch Residential Stock [11], by Leth-Petersen and Togeby for Denmark [30] and Liao and Chang for the USA [31]. The decrease in energy consumption in newer buildings is usually attributed to the improved insulation and the use of more efficient lighting and air conditioning systems [31,32]. However, some earlier studies have observed an increase in household electricity consumption of new buildings due to increased use of air conditioning and other high-consumption appliances [33], while others report no effect [34].

The presence of energy efficient materials in the building structure has also been associated with the energy consumption [20,35]. Insulation of the façade, followed by double glazing, insulation on the ground, and insulation of the windows was found to reduce energy use in Guerra Santin's study [11]. Similar results have been reported by Hirst and Goeltz [22].

2.2. Impact of socio-demographic characteristics

The relationship between energy consumption and household social and economic characteristics has been extensively studied. The most commonly used socio-economic characteristics are the household's size and composition and householders' age, income, education level, and employment status.

The household size, i.e. the number of persons in the household, is positively correlated with energy consumption and considered as one of the most influencing socio-demographic factors [12,16,21,31,36]. The energy consumption varies further with household composition. As reported by Baker and colleagues [37], the presence of children or elderly people in the household is associated with increased occupancy and more time spent at home and thus may increase energy consumption. Similar results have been reported by Van Raaij and Verhallen [38], while according to Hirst et al. [28], the influence of the number of adults is larger than the influence of the number of children. Furthermore, studies have shown that the per capita consumption is lower for larger than smaller households, probably due to the shared used of equipment [35,39].

The age of occupants is an important variable, however the most commonly referred variable is the age of the householder or responder and it is used as a proxy for the life-cycle of a particular household and the time people spend at home [15,36]. The age of the householder usually has similar effect as the household size on the energy requirements in different countries [40]. A significant relation between the age of the householder and the energy requirements has been reported in numerous studies [7,11,19,31,35,36].

Household income is positively associated with the energy consumption and this is consistent in the majority of studies [5,11,12,28,35,37,41–44]. Yet, Yust and colleagues suggest that household income alone is not a significant determinant of energy consumption [20], and recommend to include an interaction term between household income and the size of the structure to reach to conclusions. Moreover, as reported by Santamouris and colleagues [45,46], there are complexities in the energy consumption of lower-income groups; low income households use less energy, but they are less able to reduce their energy use compared to the high-income households, partly because they are more likely to reside in older buildings with poor envelope conditions [45].

Through the literature review it became apparent that for many factors affecting residential energy consumption, results can be ambiguous, and although there are factors that certainly affect electricity consumption, there are factors as well whose effect is unclear and further research is required.

3. Data and analysis methods

The data used for this study comes from the ICE-WISH project database. The ICE-WISH project (Grant Agreement number: 270898) was an Information and Communication Technology (ICT) project funded by the European Commission aiming to improve behavior-based energy efficiency in social housing while maintaining optimum indoor environmental conditions. In total, 244 dwellings at ten country pilots (Belgium, Bulgaria, Denmark, France, Germany, Greece, Italy, Poland, Spain and the UK) were involved in the project. This distribution reflects not only different climatic conditions, but also different cultures and social housing ownership structures.

3.1. Procedure and material

The database consists of data from measured utilities matched with data from quantitative surveys. To fulfill the quantitative data requirements two different questionnaires were developed: a building and systems questionnaire for the housing managers and a questionnaire for the housing occupants.

The building and systems questionnaire consisted of several sections, each devoted to different aspects of operation, maintenance, monitoring, and construction of the building.

The housing occupant questionnaire consisted of sections concerning basic socio-demographic and socio-economic characteristics of the occupants, such as age, household size, income, education and employment status.

Table 1
Electricity consumption and energy consumption for SWH: whole sample and groups (mean, minimum and maximum values).

Annual Electricity Consumption (kWh)	Mean (SD)	Min	Max
Whole sample	2302 (1147)	720	7619
Low group	1293 (240)	720	1644
Medium group	2094 (249)	1675	2543
High group	3581 (1117)	2576	7619
Annual energy consumption for SWH (kWh)	Mean (SD)	Min	Max
Whole sample	4063(2124)	237	9396
Low group	1642(612)	237	2848
Medium group	4064(728)	2933	5296
High group	6482(925)	5304	9396

3.2. Sample of participants

The study took place in 2011. A total of 218 households completed the housing occupant questionnaire. From these households, energy consumption for space and water heating was available only for 144 dwellings, thus the final data set used for this study consisted of 144 records. Moreover, only 34% of the participants responded to the question concerning income, therefore we didn't include income as a variable in our analysis. The average age of the respondents was 38 years (SD = 19.6); for the number of occupants that defined their gender, the number of female occupants is either greater (Bulgaria, Denmark, France, Germany, Poland, Spain) or equal (Belgium, Italy) to the number of male occupants in each pilot.

Buildings in our study are equipped with a central SWH system (gas and district heating). The only exception is a small number of dwellings, all located in the same pilot building in Bilbao, Spain. The building was built in 1919 and thus these dwellings have no heating system at all and use electricity for hot water. Hence, in our sample the heating has no impact on electricity consumption and allows for separately investigating drivers of high electricity consumption and high consumption for space and water heating. Moreover all dwellings are apartments, so we do not consider dwelling type as a variable.

The average annual electricity consumption for the participating households is 2302 kWh (SD = 1147), while the average annual energy consumption for SWH is 4063 kWh (SD = 2124) (Table 1). These values are in general lower than the European annual average electricity consumption for lighting and appliances and the energy consumption for space heating which are 2266 kWh and 10,576 kWh, respectively (reference year 2011) [47].

The available data were stratified into three equally sized groups, following the method of stratification previously used by Jones et al. [34], Firth et al. [48], and Summerfield et al. [49,50]. The 48 households with the lowest electricity consumption were classified as the "low electricity group", the middle 48 as the "medium electricity group", and the 48 households with the highest electricity consumption as the "high electricity group". We used the same stratification method for the energy consumption for SWH, to classify the low SWH group, the medium SWH group and the high SWH group, respectively. Because our goal was to study the effects of socio-demographic and building characteristics on the high electricity and high energy consumption for SWH groups, we then merged the low and the medium consumption groups into one.

3.3. Analysis methods

Odds ratio (OR) analysis was performed to examine the impact of specific factors on the electricity consumption and the consumption for SWH of the households. These factors involve the socio-demographic characteristics of the occupants and the building characteristics. This method has been also used by Jones and Lomas [34] in a study of the socio-economic and dwelling characteristics affecting high electricity demand in UK homes.

Odds are a different expression of the probability: the odds of an event happening is the probability that the event will happen divided by the probability that the event will not happen [51,52]. The odds ratio is the odds of an outcome in one group divided by the odds of the outcome in the other group and is a statistic used to assess the risk of a particular outcome if a certain factor is present.

An OR value of 1 indicates that the factor does not affect the odds of outcome. An OR value greater than 1 indicates that the factor is associated with higher odds of outcome, whereas an OR value less than 1 indicates that the factor is associated with lower odds of outcome.

In our analysis the outcome of interest is high electricity consumption (Table 2) and high energy consumption for SWH (Table 3), analyzed against the factors of interest, e.g. the socio-demographic and building characteristics which are listed in Tables 2 and 3. For a given factor, the OR is the number of households with high electricity consumption divided by the number of households with low-medium electricity consumption, divided by the same ratio for the reference group. For each factor, the reference group is selected based on two criteria (i) the household did not have the factor, or (ii) the factor represented the majority of the sample [34].

For example, the odds ratio for being in the high electricity group given the presence of children less than 5 years old is:

$$\mathrm{OR} = \frac{a/c}{b/d} = \frac{11/18}{35/80} = 1.40$$

where a is the number of households with children less than 5 years old in the high electricity consumption group; b is the number of households with no children less than 5 years old in the high electricity consumption group; c is the number of households with children less than 5 years old in the low-medium consumption group; d is the number of households with no children in the high electricity consumption group.

The 95% confidence intervals (CI) for the OR are calculated using the following equation:

$$CI_{\pm 95\%} = e^{ln(OR)\pm\sqrt{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}}} \quad (1)$$

4. Results

The socio-demographic characteristics that were examined in our study were: the number of occupants, the presence of children less than four years old, the presence of children between 5 and 15 years old, the age of the Household Responder (HR), the employment status of the HR, and the education level of the HR.

The building characteristics that were examined were: the period during which the building was built, the number of bedrooms, the total floor area, the presence of electric water heating, and the U-value of external walls.

The impact that these characteristics may have on high electricity consumption and high energy consumption for space and water heating is discussed in this section.

4.1. Socio-demographic characteristics

Regarding the impact of socio-demographic variables on electricity consumption, we found that the main determinant of high electricity use is the household size. The odds of being in the high electricity consumption group (HEC) group is 1.78, 5.60, 4.11 and

Table 2
Odds ratio results for the socio-economic and building characteristics affecting high electricity consumption.

Characteristics	Low-medium group	High group	Odds ratio	(95% CI)
Socio-demographic characteristics				
Number of occupants				
1	30	5	Reference	
2	27	8	1.78	(0.52,6.10)
3	15	14	**5.6**	(1.70,18.49)***
4	19	13	**4.11**	(1.26,13.37)*
5+	7	6	**5.14**	(1.21,21.80)*
Children				
No children less than 5 years old	80	35	Reference	
Children less than 5 years old	18	11	1.4	(0.60,3.26)
No children between 5 and 15 years old	58	23	Reference	
Children between 5 and 15 years old	30	18	1.51	(0.71,3.23)
Age of householder				
<35	45	26	Reference	
36–50	5	6	2.08	(0.58,7.48)
51–65	17	6	0.61	(0.21,1.74)
>65	13	2	0.27	(0.06,1.27)
Employment status of HR				
Employed (full time or part time)	49	27	Reference	
Unemployed	7	6	1.56	(0.47,5.10)
Retired	29	7	0.44	(0.17,1.13)
Sick-Injured	9	3	0.6	(0.15,2.43)
Students	4	3	1.36	(0.28,6.53)
Education level				
Secondary	25	11	Reference	
Less than Secondary	13	14	2.45	(0.87,6.90)
University or more	13	7	1.22	(0.38,3.91)
Building characteristics				
Year built				
<1970	23	8	0.94	(0.36,2.44)
1971–2000	21	18	**2.31**	(1.03,5.21)**
>2000	54	20	Reference	
Number of bedrooms				
1	30	3	Reference	
2	55	31	**5.64**	(1.59,19.99)***
3 or more	13	12	**9.23**	(2.23,38.29)***
Total floor area				
<60 m^2	45	11	Reference	
60–80 m^2	38	24	**2.58**	(1.12,5.95)**
>80 m^2	15	11	**3**	(1.08,8.32)**
Electric water heating				
None	96	38	Reference	
Electric water heater	2	8	**10.11**	(2.05,49.77)***

* Significant at the 10% level.
** Significant at the 5% level.
*** Significant at the 1% level.

5.14 higher in households with two, three, four and more than five persons respectively, compared to single households. The odds of households with two occupants does not reach statistical significance, however, all other odds were more than twice that of households with two occupants and were statistically significant. As expected, there is a positive association between the number of occupants and the electricity consumption. Our findings are consistent with a wide number of previous studies [7,12,49,53,54].

Although significantly increased electricity consumption has been observed in homes with children and teenagers in previous studies [12,54,55], no such effect was found in our study. The odds of being in the HEC group is 1.40 higher for families with children less than five years old compared to families with no children in these ages, and 1.51 for families with children between 5 and 15 years old compared to families with no children in this range. However, these increased odds do not reach statistical significance. Similarly, Huebner et al [56], found no effect of teenagers on electricity consumption, and attribute this finding to differences in samples and dependent variables.

Considering the existence of children as a binary factor (children/no children), without specifying for differences in the number or the age of the children, we found similar results. The odds ratio is 1.45 and is not statistically significant.

Regarding the age of the HR, previous studies indicate that dwellings with HR over 65 years of age are significantly less likely to be high electricity consumers [11,23,27,34]. Results in our study indicate a similar effect, but the decreased odds ratio does not reach statistical significance. More specifically, the odds of being in the high HEC group is 2.08 higher for householder's age between 36 and 50 years compared to householders less than 36 years old (reference group). As age increases, the odds of being in the high electricity grouped are inversed, i.e., the odds of being in the high electricity consumption group is 0.61 and 0.27 lower for householder's age between 51 and 65 years old and more than 65 years old respectively, compared to the reference group. Since the 95%

Table 3
Odds ratio results for the socio-economic and building characteristics affecting high energy consumption for SWH.

Characteristics	Low-medium group	High group	Odds ratio	(95% CI)
Socio-demographic characteristics				
Number of occupants				
1	24	11	Reference	
2	21	14	1.45	(0.54,3.89)
3	19	10	1.15	(0.40,3.27)
4	25	7	0.61	(0.20,1.84)
5+	7	6	1.87	(0.51,6.88)
Children				
No children less than 5 years old	73	42	Reference	
Children less than 5 years old	23	6	0.45	(0.17,1.20)
No children between 5 and 15 years old	61	35	Reference	
Children between 5 and 15 years old	35	13	0.65	(0.30,1.38)
Age of householder				
<35	52	19	Reference	
36–50	5	6	3.28	(0.90,12.03)
51–65	19	4	0.58	(0.17,1.91)
>65	10	5	1.37	(0.41,4.52)
Employment status				
Employed (full time or part time)	52	24	Reference	
Unemployed	7	6	1.86	(0.56,6.12)
Retired	23	13	1.22	(0.53,2.82)
Sick-Injured	10	2	0.43	(0.09,2.13)
Students	4	3	1.63	(0.34,7.84)
Education level				
Secondary	18	18	Reference	
Less than secondary	16	11	0.69	(0,25,1.88)
University or more	14	6	0.43	(0.13,1.36)
Building characteristics				
Year built				
<1970	16	15	**3.69**	(1.49,9.11)***
1971–2000	21	18	**3.37**	(1.45,7.86)***
>2000	59	15	Reference	
Number of bedrooms				
1	23	10	Reference	
2	57	29	1.17	(0.49,2.78)
3 or more	16	9	1.29	(0.43,3.90)
Total floor area				
<60 m^2	45	11	Reference	
60–80 m^2	39	23	**2.41**	(1.04,5.57)**
>80 m^2	12	14	**4.77**	(1.73,13.16)***
Walls U-value				
Low	30	0	Reference	
Medium	26	4	**1.15**+	(1.00,1.33)*
High	40	44	**2.10**+	(1.68,2.63)***
Electric water heating				
None	88	46	Reference	
Electric water heater	8	2	0.48	(0.10,2.35)

*Significant at the 10% level.
**Significant at the 5% level.
***Significant at the 1% level.
+Reported values are risk factors, because the number of households with low *U*-values in high SWH group is zero.

CI of these odds spans 1, neither the increased nor the decreased odds reach statistical significance.

The employment status of the HR had no influence on the probability of high electricity use. Compared to households with employed occupants, the odds of being in the HEC group are 0.48 lower for households with retired occupants, while the relevant odds for households with unemployed occupants and students are 1.56 and 1.36, respectively. Jones and Lomas [34] found that households with retired HR's were significantly less likely to be high consumers than those with an employed HR. Likewise, the relevant odds ratio in our analysis, indicate a similar effect, but not statistically significant.

In accordance with some previous studies [9,34,41], we found no association between the education level of the HR and high electricity consumption. The odds of being in the HEC group are 2.45 higher for HR with less than secondary education compared to HR with secondary education, but the effect is not significant.

4.2. Dwelling characteristics

The dwelling's characteristics, concerning their size, namely the number of bedrooms and the total floor area, were found to have an effect on the likelihood of a house to belong in the HEC group.

In accordance with previous studies, we found that the number of bedrooms in the dwelling has a significant effect on high electricity consumption [53,57,58]. Compared to dwellings with only one bedroom, the odds of being in the HEC group is 5.6 for dwellings with two bedrooms, and 9.2 for dwellings with three or more bedrooms. These odds are all statistically significant.

Similarly, dwellings with bigger floor area were significantly more likely to be high electricity consumers [7,49,53,54]. Com-

pared to dwellings with floor area of less than 60m^2, the odds of being in the HEC group is 2.58, for dwellings with a floor area between 60 and 80 m^2, and 3 for dwellings with floor area greater than 80 m^2.

The period in which the dwelling was built was found to influence the probability of the occupying household being in the HEC group. Compared to buildings built after 2000, the odds of being in the HEC for buildings built between years 1970 and 2000 is 2.31. This finding is consistent with earlier studies [12,59], that attribute higher electricity in older homes to increased heat losses, less insulation and less efficient appliances. However, there are studies that report no significant effects of building age on electricity consumption [7,34,56].

Not surprisingly, dwellings in which electricity is the primary source of energy for hot water are significantly more likely to be high electricity consumers, which is in line with findings from other studies [7,34]. For these dwellings, the odds of being in the HEC group is 10.11 compared to dwellings that use gas.

4.3. Energy use for space and water heating

The OR analysis indicates that the examined socio-demographic characteristics have no effect on high energy consumption for SWH.

Neither the household size, nor the presence of children and the age of HR appeared to affect the probability of a household to consume more energy for SWH. Similarly, the employment status and the education level have no effect on high energy consumption for SWH. As shown in Table 3, for all the characteristics the confidence internals span unity.

The finding of socio-demographic characteristics appearing to have no statistically significant effect in energy consumption for SWH, is consistent with previous studies analyzing disaggregated data [60–62], where household characteristics turn out to not explain energy consumption for heating in a satisfactory manner, and the influence of HR age and employment status is inconclusive. However, there are studies that report positive correlation between energy consumption for space heating and age of householder [63,64], and attribute higher elderly household consumption to the intensity of use or occupation and the preference for higher indoor air temperatures. On the contrary, with the exception of the number of bedrooms and the electric water heating, all the examined dwelling characteristics have a statistical significant effect on high energy use for SWH.

Compared to dwellings with floor area of less than 60 m^2, the odds of being in the high SWH consumption group is 2.41, for dwellings with a floor area between 60 and 80 m^2, and 4.77 for dwellings with floor area greater than 80 m^2. Unsurprisingly, previous studies [11,12,14,17,21] agree that there is a significant and positive effect of dwelling floor area on the energy consumption for space heating, as bigger floor area means greater demand for space heating. However, compared to dwellings with only one bedroom, the odds of being in the high consumption for SWH group is 1.17 for dwellings with two bedrooms, and 1.29 for dwellings with three or more bedrooms. Thus, the number of bedrooms has no effect on high SWH group.

The period in which the dwelling was built was found to influence the probability of the occupying household being in the group with high energy consumption for SWH. Compared to buildings built after 2000, the odds of being in the high SWH group for buildings built prior to 1970 is 3.69, and for buildings built between 1970 and 2000 is 3.37. Similar results were found by Guerra Santin et al. [11], Brounen et al. [12] and Leth-Petersen and Togeby [30].

Risk values for the *U*-value of external walls indicate that dwellings whose external walls are characterized by higher *U*-values are significantly more likely to belong in the high SWH consumption group. Specifically, the probability of being in the high SWH consumption group is 1.15 and 2.10 higher for dwellings with medium and high *U*-value external walls, respectively, compared to dwellings with external walls of low *U*-value. These expected results are in line with previous research [12,30].

5. Discussion

Our analysis focused on examining factors that affect high electricity consumption and high energy consumption for space and water heating in social housing. The analysis showed that non-heating electricity consumption is impacted by both socio-demographic and building related variables, while high energy consumption for SWH is mainly associated with building related variables.

5.1. Socio-demographic characteristics

Previous studies have indicated a significant positive effect of dwelling factors such as dwelling age, number of rooms, number of bedrooms, and total floor area on domestic electricity use. Compared to studies that focus on electricity consumption, studies examining the effect of socio-demographic factors on energy consumption for SWH are less often found in literature. The effect of building characteristics, physical or technical, on energy consumption for SWH is much more studied and explained, and scholars seem to agree on their influence.

Our findings regarding the socio-demographic predictors of energy use are summarized as follows:

The number of occupants tends to be positively associated with high electricity consumption. This is a finding consistent with previous research [7,12,49,53,54], suggesting that there is a positive, significant relationship between the household size and the electricity use, such that larger households consume more electricity compared to smaller households. Such an effect is usually attributed to the possession of more appliances and the more intensive use of some of these appliances, as larger households have greater energy requirements for cooking, cleaning and washing.

However, the number of occupants has no statistically significant effect on energy consumption for SWH. In line with previous studies [60,61], we found no evidence of higher energy consumption for SWH for households with additional members. Although water heating can be considered quite dependent on the number of occupants, it seems that total heating requirements are the dominant factor, and the fact that dwellings still need to be heated, either they are occupied by one or more members. Yet, there are studies reporting a positive correlation between household size and energy consumption for SWH. However, caution is required in the interpretation of these results [12], because it could be that the true determinant is the dwelling size: bigger households choose bigger dwellings, which in terms affects household's requirements for space heating.

The HR age was found to have no effect on the probability of a household being a high energy consumer of either electricity or energy consumption for SWH. In fact, relevant literature is inconclusive as far as it concerns HR age, and show a mix of effects or no effect at all. Some studies [11,23,27,34] support that older people consume less electricity and attribute this finding to lower income, smaller number of household occupants and possession of a smaller number of appliances. Other studies [63,64] support a positive association between energy consumption for space heating and HR age, attributing the effect to the fact that elderly people may leave in older and less insulated and efficient buildings, spend more time at home and prefer higher indoor air temperature to be

comfortable. Actually, our analysis, in agreement with some previous studies [7,12,53], indicates a non monotonic relationship with age, where households with a HR age between 35 and 50 years old are more likely to be high energy consumers.

In line with previous research [9,34,41,53], neither the employment status, nor the education level of the HR was found to have an effect on household's energy consumption. Employment status is often used as a proxy for disposable income or occupancy, i.e. full time employees can spend more on every day energy use and appliances and invest more in energy efficiency measures, while spend fewer hours at home compared to unemployed or retired householders. However, these effects cannot be consistently observed or measured, and thus their influence is rather uncertain and cannot explain substantial variability in energy consumption. Similarly, education is used as a proxy for pro-environmental behavior, and can be associated with increased knowledge and awareness, but most of the studies have found that there is no significant direct impact on household energy consumption.

5.2. Dwelling characteristics

The period during which the building was built has a significant effect on high energy consumption. In consistency with previous research [12,18,20,28–31,59], we found that electricity and energy consumption for SWH increase with the age of the building. These results suggest that the improvement of building standards and the energy labeling of appliances have resulted in better constructed and more efficient buildings, equipped with more efficient appliances and lighting systems.

Similarly, the total floor area had a significant positive effect on high energy consumption. We found that buildings with a floor area greater than 80 m^2 were much more likely to be high energy consumers than those with a floor area smaller than 60 m^2. This finding is in line with previous studies [7,11,12,14,17,21,34,49,53,54] which consistently report a positive strong total floor area effect, with large dwellings using both more electricity and more energy for SWH.

The number of bedrooms was found to have a significant positive effect only on high electricity consumption. Previous research has found similar results [53,57,58], and the higher electricity use in dwellings with more bedrooms is usually attributed to the use of more appliances, lighting and maybe portable heaters. The indication of a non significant relationship between the number of bedrooms and energy consumption for space heating may be attributed and explained by the fact that it is the entire dwelling that is heated, rather than only specific rooms.

Dwellings in which electricity is used for hot water were significantly more likely to be high electricity consumers, and previous [7,34] studies agree that there is a significant and positive effect of an electric water heating system on electricity use. Similarly, literature [11,12] constantly agrees that external wall *U*-value has a significant and positive effect on energy use for SWH. Both these results are expected and trivial to understand. As water heating represents a basic energy use in a dwelling, if it is provided by electricity, the electricity use will be increased. Likewise, when dwellings external walls *U*-values are high, energy consumption for space heating will be increased, as less insulation increases heat loss.

6. Conclusions

The objective of this study was to identify the major determinants of high energy use in social housing. To this effect, odds ratio analysis was used to distinctly investigate the effects of the socio-demographic and building characteristics on the electrical energy consumption and the consumption for space and water heating. Results indicate that high electricity consumption and high energy consumption for space and water heating in households are related to different factors.

High electricity consumption in social housing is related to both socio-demographic characteristics of the occupants and characteristics of the buildings in which they live. The household size, the number of bedrooms, the total floor area, the age of the building and the existence of electric water heating are the main factors affecting high electricity consumption. However, the employment status, the education level, the presence of children and technical buildings characteristics, such as external wall *U*-values have no statistically significant effect on high electricity consumption.

High energy consumption for space and water heating, in the contrary, is only related to building characteristics, including size, year of construction and external wall *U*-values, whereas occupants' socio-demographic characteristics are shown to have no direct effect.

Overall, the odds ratio analysis suggests that households with more occupants, living in older and less insulated buildings, with greater floor area and electric water heating, are more likely to be among the high energy consumers. Employment status and education level appear to have no effect on electricity, nor on energy consumption for space and water heating.

The outcomes of this paper have implications for research and policy. With approximately 20% of greenhouse gas emissions attributed to the residential sector, advancing our understanding of the factors that affect residential consumption is important. While improving energy efficiency through technological advances in appliances, energy systems and building's physical and technical structure has been for long a priority, household's socio-demographic characteristics have not been a bid priority until recent years; these need to be investigated and incorporated into future energy policy.

Acknowledgments

This research was supported in parts by the European Funded Project ICE-WISH (Grant Agreement number: 270898).

The authors would like to thank the participants of the ICE-WISH project: Alain Lusardi and Giancarlo Tofanelli from Consorzio Nazionale Casa Qualità (Casa Qualita) in Italy; Bernard Wallyn, Kristien Van Hemelryck, Philippe Plaghki and Hannes Benaets from Vlaamse Maatschappij voor Sociaal Wonen (VMSW) in Belgium; Eleonora Gaydarova from the Union of Homeowners Associations(CAC) in Bulgaria; Rie Oehlenschlaeger, Lulu Grønlund and Palle Jørgensen from Boligforeningen Ringgården (RING) in Denmark; Nitschke Juergen, Siebert Julia and Wolfgang Pfeuffer from Joseph Stiftung in Germany; Rosario Vallejo Ilarduia, Monike Nicolas,Oihana Hernaez and Eugenia Atin from Viviendas Municipales deBilbao in Spain (Bilbao Viviendas); Mike Roys, Ranjit Bassi, Kim-Noonan and Lorna Hamilton from the Building Research Establishment (BRE) in the United Kingdom.

References

[1] W. Abrahamse, L. Steg, C. Vlek, T. Rothengatter, The effect of tailored information, goal setting, and tailored feedback on household energy use, energy-related behaviors, and behavioral antecedents, J. Environ. Psychol. 27 (2007) 265–276.
[2] http://ec.europa.eu/eurostat/documents/3217494/6898731/KS-EN-15-001-EN-N.pdf/e5851c73-9259-462e-befc-6d037dc8216a.
[3] https://www.eia.gov/totalenergy/data/annual/#consumption.
[4] http://www.eea.europa.eu/data-and-maps/indicators/energy-efficiency-and-energy-consumption-5/assessment.
[5] R. Haas, Energy efficiency indicators in the residential sector: what do we know and what has to be ensured? Distribution 25 (1997) 789–802 PII:S0301-4215(97)00069-4.
[6] W. Abrahamse, L. Steg, C. Vlek, T. Rothengatter, A review of intervention studies aimed at the household energy conservation, J. Environ. Psychol. 25 (3) (2005) 279–291.

[7] A. Kavousian, R. Rajagopal, M. Fischer, Determinants of residential electricity consumption: using smart meter data to examine the effect of climate, building characteristics, appliance stock, and occupants' behavior, Energy 55 (2013) 184–194.
[8] R.V. Jones, A. Fuertes, K. Lomas, The socio-economic, dwelling and appliance related factors affecting electricity consumption in domestic buildings, Renew. Sustain. Energy Rev. 43 (2015) 901–917.
[9] J.M. Cayla, N. Maizi, C. Marchand, The role of income in energy consumption behaviour: evidence from French households data, Energy Policy 39 (12) (2011) 7874–7883.
[10] B. Sütterlin, T.A. Brunner, M. Siegrist, Who puts the most energy into energy conservation? A segmentation of energy consumers based on energy-related behavioral characteristics, Energy Policy 39 (12) (2011) 8137–8152.
[11] O. Guerra Santin, L. Itard, H. Visscher, The effect of occupancy and building characteristics on energy use for space and water heating in Dutch residential stock, Energy Build. 41 (11) (2009) 1223–1232.
[12] D. Brounen, N. Kok, J.M. Quigley, Residential energy use and conservation: economics and demographics, Eur. Econ. Rev. 56 (2012) 931–945.
[13] A. Druckman, T. Jackson, Household energy consumption in the UK: a highly geographically and socio-economically disaggregated model, Energy Policy 36 (8) (2008) 3177–3192.
[14] R. Ewing, F. Rong, in: The Impact of Urban Form on US Residential Energy Use, Housing Policy Debate, 2008, pp. 1–30.
[15] L. Schipper, D. Hawk, E. Vine, S. Bartlett, Linking life-styles and energy use: a matter of time? Annu. Rev. Energy 14 (1989) 273–320.
[16] S. Kelly, Do homes that are more energy efficient consume less energy? A structural equation model of the English residential sector, Energy 36 (9) (2011) 5610–5620.
[17] Y. Shimoda, A. Takahiro, T. Ayako, M. Minoru, Evaluation of city-scale impact of residential energy conservation measures using the detailed end-use simulation model, Energy (2007) 1617–1633.
[18] R. Nesbakken, Energy consumption for space heating: a discrete–continuous approach, Scand. J. Econ. 103 (1) (2001) 165–184.
[19] J.R.B. Ritchie, H.G.M. Gordon, J.D. Claxton, Complexities of household energy consumption and conservation, J. Consum. Res. 8 (3) (1981) 233–242.
[20] B.L. Yust, D.A. Guerin, J.G. Coopet, Residential energy consumption: 1987 to 1997, Fam. Consum. Sci. Res. J. 30 (3) (2002) 323–349.
[21] N. Kaza, Understanding the spectrum of residential energy consumption: a quantile regression approach, Energy Policy 38 (11) (2010) 6574–6585.
[22] R. Halvorsen, Residential demand for electric energy, Rev. Econ. Stat. 57 (1) (1975) 12–18.
[23] E. Hirst, A model of residential energy use, Simulation 30 (3) (1978) 69–74.
[24] H. Houthakker, Residential electricity revisited, Energy 1 (1) (1980) 29–41.
[25] J.A. Dubin, D. Mcfadden, An econometric analysis of residential electric appliance holdings and consumption, Society 52 (2) (1984) 345–362.
[26] R. Haas, L. Schipper, Residential energy demand in OECD-countries and the role of irreversible efficiency improvements, Energy Econ. 20 (4) (1998) 421–442.
[27] B. Baltagi, Comparison of forecast performance for homogeneous, heterogeneous and shrinkage estimators: some empirical evidence from US electricity and natural-gas consumption, Econ. Lett. 76 (3) (2002) 375–382.
[28] E. Hirst, R. Goeltz, J. Carney, Residential energy use: analysis of disaggregate data, Energy Econ. (1982) 74–82.
[29] S. Chen, H. Yoshino, N. Li, Statistical analyses on summer energy consumption characteristics of residential buildings in some cities of China, Energy Build. 42 (1) (2010) 136–146.
[30] S. Leth-Petersen, M. Togeby, Demand for space heating in apartment blocks: measuring effect of policy measures aiming at reducing energy consumption, Energy Econ. 23 (2001) 387–403.
[31] H.C. Liao, T.F. Chang, Space-heating and water-heating energy demands of the aged in the US, Energy Econ. 24 (2002) 267–284.
[32] A.L. Linden, A. Carlssonkanyama, B. Eriksson, Efficient and inefficient aspects of residential energy behavior: what are the policy instruments for change? Energy Policy 34 (14) (2006) 1918–1927.
[33] US Energy Information Administration, US Household Electricity Report 2005, US EIA, Washington, DC, 2005 Tech. Rep.;. URL http://www.eia.gov/emeu/reps/enduse/er_contents.html.
[34] R.V. Jones, K.J. Lomas, Determinants of high electrical energy demand in Uk homes: Socio-economic and dwelling characteristics, Energy Build. 101 (2015) 24–34.
[35] V.I. Ugursal, A.S. Fung, Impact of appliance efficiency and fuel substitution on residential end-use energy consumption in Canada, Energy Build. 24 (2) (1996) 137–146 1996.
[36] B.C. O'Neill, B.S. Chen, Demographic determinants of household energy use in the United States, Popul. Dev. Rev. 28 (2002) 53–88.
[37] P. Baker, R. Blundell, J. Micklewright, Modelling household energy expenditures using micro-data, Econ. J. 99 (397) (1989) 720–738.
[38] W.F. Van Raaij, T.M.M. Verhallen, A behavioral model of residential energy use, J. Econ. Psychol. (1983) 39–63.
[39] W.K. Fong, R. Kimura, Y.F. Lun, H. Matsumoto, Influences of indirect lifestyle aspects and climate on household energy consumption, J. Asian Archit. Build. Eng. 6 (2) (2007) 395–402.
[40] W. Lenzen, H. Cohen, S. Pachauri, A comperative multivariate analysis of household energy requirements in Australia, Brazil, Denmark, India and Japan, Energy 31 (2006) 181–207.
[41] B. Gatersleben, L. Steg, C. Vlek, Measurement and determinants of environmentally significant consumer behavior, Environ. Behav. 34 (2002) 335–362.
[42] J.O. Jensen, Measuring consumption in households: interpretations and strategies, Ecol. Econ. 68 (1–2) (2008) 353–361.
[43] L. Lutzenhiser, M.H. Gossard, Lifestyle, status, and energy consumption, Paper Presented at the ACEEE Summer Study on Energy Efficiency in Buildings, 2000.
[44] G. Brandon, A. Lewis, Reducing household energy consumption: a qualitative and quantitative filed study, J. Environ. Psychol. 19 (1) (1999) 75–85.
[45] M. Santamouris, K. Kapsis, D. Korres, I. Livada, C. Pavlou, M.N. Assimakopoulos, On the relation between the energy and social characteristics of the residential sector, Energy Build. 39 (8) (2007) 893–905.
[46] M. Santamouris, S.M. Alevizos, L. Aslanoglou, D. Mantzios, P. Milonas, I. Sarelli, S. Karatasou, K. Cartalis, J.A. Paravantis, Freezing the poor—Indoor environmental quality in low and very low income households during the winter period in Athens, Energy Build. 70 (2014) 61–70.
[47] http://www.indicators.odyssee-mure.eu/online-indicators.html.
[48] S. Firth *, K. Lomas, A. Wright, R. Wall, Identifying trends in the use of domestic appliances from households electricity consumption measurements, Energy Build. 40 (5) (2008) 926–936.
[49] A.J. Summerfield, R.J. Lowe, H.R. Bruhns, J.A. Caeiro, J.P. Steadman, T. Oreszczyn, Milton Keynes Energy Park revisited: changes in internal temperatures and energy usage, Energy Build. 39 (7) (2007) 783–791.
[50] A.J. Summerfield, A. Pathan, R.J. Lowe, T. Oreszczyn, Changes in energy demand from low-energy homes, Build. Res. Inf. 38 (1) (2010) 42–49.
[51] D. Grimes, K.F. Schulz, Making sense of odds and odds ratios, Obstet. Gynecol. 111 (2) (2008).
[52] J.M. Bland, D.G Altman, The odds ratio, Br. Med. J. 320 (2000) 1468.
[53] Y.G. Yohanis, J.D. Mondol, A. Wright, B. Norton, Real-life energy use in the UK: how occupancy and dwelling characteristics affect domestic electricity use, Energy Build. 40 (6) (2008) 1053–1059.
[54] F. Bartiaux, K. Gram-Hanssen, Socio-political factors influencing household electricity consumption: a comparison between Denmark and Belgium, in: What Works & Who Delivers?: Proceedings of the ECEEE 2005 SummerStudy, European Council for an Energy Efficient Economy, 2005, pp. 1313–1325. [online].
[55] F. McLoughlin, A. Duffy, M. Conlon, Characterising domestic electricity consumption patterns by dwelling and occupant socio-economic variables: an Irish case study, Energy Build. 48 (2012) 240–248.
[56] G. Huebner, D. Shipworth, I. Hamilton, A. Chalabi, T. Oreszczyn, Understanding electricity consumption: a comparative contribution of building factors, socio-demographics, appliances, behaviours and attitudes, Appl. Energy 177 (2016) 692–702.
[57] M. Bedir, E. Hasselaar, L. Itard, Determinants of electricity consumption in Dutchdwellings, Energy Build. 58 (2013) 194–207.
[58] I.G. Hamilton, P.J. Steadman, H. Bruhns, A.J. Summerfield, R. Lowe, Energy efficiency in the British housing stock: energy demand and the Homes Energy Efficiency Database, Energy Policy 60 (2013) 462–480.
[59] E. Leahy, S. Lyons, Energy use and appliance ownership in Ireland, Energy Policy 38 (8) (2010) 4265–4279.
[60] G. Capper, A. Scott, The economics of house heating: further findings, Energy Econ. 4 (2) (1982) 134–138.
[61] A. Schuler, C. Weber, U. Fahl, Energy consumption for space heating of West Germany households: empirical evidence, scenario projections and policy implications, Energy Policy 28 (12) (2000) 877–894.
[62] R.C. Sonderegger, Movers and stayers: the resident's contribution to variation across houses in energy consumption for space heating, Energy Build. 1 (3) (1978) 313–324.
[63] E. Yamasaki, N. Tominaga, Evolution of an aging society and effect on residential energy demand, Energy Policy 25 (11) (1997) 903–912.
[64] H. Liao, T. Chang, Space-heating and water-heating energy demands of the aged in the US, Energy Econ. 24 (3) (2002) 267–284.